An $\mathcal{O}$-adapted slice filtration and detection of slice connectivity by geometric fixed points

Abstract

Fix a finite group G and let $\mathcal{O}$ be an *incomplete transfer system*, for example the transfer data encoded by a G–N_∞ operad. We define an $\mathcal{O}$-adapted slice filtration on an $\mathcal{O}$-equivariant stable category and prove that for a *connective* $\mathcal{O}$-spectrum X, $\mathcal{O}$-slice connectivity is detected by the ordinary connectivity of geometric fixed points. The detecting function is the maximal size of an $\mathcal{O}$-admissible transitive orbit in each subgroup.

Contents

1 Transfer systems and the $\mathcal{O}$-stable category

Fix a finite group G.

Definition 1.1 (Transfer system)**.** A *transfer system* (or *incomplete transfer system*) on G is a relation $\leq_{\mathcal{O}}$ on subgroups of G satisfying:

1. If $K \leq_{\mathcal{O}} H$, then $K \leq H$.
2. Reflexivity and transitivity.
3. Conjugation invariance: $K \leq_{\mathcal{O}} H \Rightarrow gKg^{-1} \leq_{\mathcal{O}} gHg^{-1}$.
4. Restriction: $K \leq_{\mathcal{O}} H$ and $L \leq H$ imply $K \cap L \leq_{\mathcal{O}} L$.

The system is *complete* if $K \leq_{\mathcal{O}} H$ for all inclusions $K \leq H$.

Transfer systems are equivalent (via work of Blumberg–Hill and others) to the admissible transitive sets appearing in an N_∞ operad and hence specify which norm/transfer maps are available [**?**, **?**].

Remark 1.2 (Ambient category)**.** For the filtration we assume an $\mathcal{O}$-equivariant stable category $\mathrm{Sp}^G_{\mathcal{O}}$ (as in [**?**]): a stable presentable category equipped with restriction and induction functors $\mathrm{Res}^G_H \dashv \mathrm{Ind}^G_H$ for subgroups $H \leq G$, and geometric fixed points $\Phi^H \colon \mathrm{Sp}^H_{\mathcal{O}} \to \mathrm{Sp}$ satisfying the usual isotropy separation formalism (e.g. $\Phi^H(X) \simeq (\widetilde{E}\mathcal{P}_H \wedge X)^H$). All arguments below only use exactness and colimit preservation of these functors, together with isotropy separation.

Definition 1.3 (Connective objects)**.** An object $X \in \mathrm{Sp}^G_{\mathcal{O}}$ is *connective* if for every subgroup $H \leq G$, $\Phi^H(\mathrm{Res}^G_H X) \in \mathrm{Sp}_{\geq 0}$ (ordinary connective spectra).

2 The $\mathcal{O}$-adapted slice filtration

2.1 Slice cells

Let $H \leq G$ and let $K \leq_{\mathcal{O}} H$. Write $\rho_{H/K}$ for the real permutation representation on the finite H-set H/K, and $S^{\rho_{H/K}}$ for the associated representation sphere in $\mathrm{Sp}^H_{\mathcal{O}}$.

Definition 2.1 ($\mathcal{O}$-slice cells)**.** For $k \geq 0$ and $K \leq_{\mathcal{O}} H \leq G$, define the *even* and *odd* $\mathcal{O}$-slice cells in $\mathrm{Sp}^G_{\mathcal{O}}$ by

$$\Sigma^{k\rho_{H/K}} G/H_+ \coloneqq \mathrm{Ind}^G_H \left(\Sigma^{k\rho_{H/K}} S^0\right), \qquad \Sigma^{k\rho_{H/K}-1} G/H_+ \coloneqq \mathrm{Ind}^G_H \left(\Sigma^{k\rho_{H/K}-1} S^0\right).$$

Their *$\mathcal{O}$-slice dimensions* are defined by

$$\dim_{\mathcal{O}} \left(\Sigma^{k\rho_{H/K}} G/H_+\right) = k|H/K|, \qquad \dim_{\mathcal{O}} \left(\Sigma^{k\rho_{H/K}-1} G/H_+\right) = k|H/K|-1.$$

2.2 The filtration

Definition 2.2 ($\mathcal{O}$-slice filtration)**.** For $n \in \mathbb{Z}$, let $\tau^{\mathcal{O},G}_{\geq n} \subseteq \mathrm{Sp}^G_{\mathcal{O}}$ be the localizing subcategory generated by all $\mathcal{O}$-slice cells of $\mathcal{O}$-slice dimension at least n. We say X is *$\mathcal{O}$-slice n-connective* if $X \in \tau^{\mathcal{O},G}_{\geq n}$.

Remark 2.3. If $\mathcal{O}$ is complete then the admissible inclusions include $e \leq_{\mathcal{O}} H$ for all H, so $\rho_{H/e}$ is the regular representation ρ_H and one recovers the usual Hill–Hopkins–Ravenel slice cells. If $\mathcal{O}$ is minimal (only identities), then the only admissible H-orbit is H/H and the filtration degenerates to a Postnikov-type filtration.

2.3 A numerical invariant of $\mathcal{O}$

Definition 2.4 (Maximal admissible orbit size)**.** For $H \leq G$, define

$$w_{\mathcal{O}}(H) \coloneqq \max\{\, |H : K| \mid K \leq_{\mathcal{O}} H \,\}.$$

3 Functoriality and fixed-point computations

Lemma 3.1 (Restriction and induction preserve slice-connectivity)**.** *Let $H \leq G$. Then $\mathrm{Res}^G_H(\tau^{\mathcal{O},G}_{\geq n}) \subseteq \tau^{\mathcal{O}|_H,H}_{\geq n}$ and $\mathrm{Ind}^G_H(\tau^{\mathcal{O}|_H,H}_{\geq n}) \subseteq \tau^{\mathcal{O},G}_{\geq n}$.*

Proof. Both functors are exact and preserve colimits. It suffices to check on generators. Induction is immediate from the definition of the generating slice cells. For restriction, one uses the standard double-coset formula expressing $\mathrm{Res}^G_H \mathrm{Ind}^G_L$ as a coproduct of inductions from subgroups of H, together with the restriction axiom of a transfer system to see that admissibility is preserved. The $\mathcal{O}$-slice dimension does not decrease under these operations. □

Lemma 3.2 (Geometric fixed points on slice cells)**.** *Let H be a finite group and work in $\mathrm{Sp}^H_{\mathcal{O}}$.*

1. *If $L < H$ is proper and $Y \in \mathrm{Sp}^L_{\mathcal{O}}$, then $\Phi^H(\mathrm{Ind}^H_L Y) \simeq 0$.*
2. *If $K \leq_{\mathcal{O}} H$ and $k \geq 0$, then*

$$\Phi^H(\Sigma^{k\rho_{H/K}} S^0) \simeq \Sigma^k S^0, \qquad \Phi^H(\Sigma^{k\rho_{H/K}-1} S^0) \simeq \Sigma^{k-1} S^0.$$

Proof. (1) The induced object $\mathrm{Ind}^H_L Y$ is built from the orbit H/L, which has isotropy contained in proper subgroups. Smashing with $\widetilde{E}\mathcal{P}_H$ kills H/L_+, hence kills $\mathrm{Ind}^H_L Y$, and therefore its H-geometric fixed points vanish.

(2) The H-fixed subspace of the permutation representation $\rho_{H/K}$ is 1-dimensional (spanned by the sum of basis vectors), so $(k\rho_{H/K})^H \cong \mathbb{R}^k$. Geometric fixed points of a representation sphere shift by the fixed subrepresentation, giving the stated ordinary suspensions. □

Proposition 3.3 (Forward direction: slices force connectivity of Φ)**.** *Let H be a finite group and $\mathcal{O}$ a transfer system on H. If $Y \in \tau^{\mathcal{O},H}_{\geq n}$, then*

$$\Phi^H(Y) \in \mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(H))}.$$

Proof. The functor Φ^H is exact and preserves colimits, hence carries a localizing subcategory generated by a set to the localizing subcategory generated by the images of that set. Thus it suffices to compute the images of generating $\mathcal{O}$-slice cells of $\tau^{\mathcal{O},H}_{\geq n}$.

Any generator induced from a proper subgroup has vanishing H-geometric fixed points by Lemma **??**(1), hence contributes nothing. The remaining generators are $\Sigma^{k\rho_{H/K}} S^0$ and $\Sigma^{k\rho_{H/K}-1} S^0$ with $K \leq_{\mathcal{O}} H$. By Lemma **??**(2)

these map to $\Sigma^k S^0$ and $\Sigma^{k-1} S^0$. The slice-dimension condition $k|H/K| \geq n$ forces $k \geq n/|H/K|$, and among admissible K the smallest possible k occurs when $|H/K|$ is maximal, i.e. when $|H/K| = w_{\mathcal{O}}(H)$. Therefore the images generate the localizing subcategory of Sp generated by spheres $\Sigma^m S^0$ for $m \geq \mathrm{ceil}(n/w_{\mathcal{O}}(H))$, which is precisely $\mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(H))}$. □

4 Geometric objects and slice connectivity

Lemma 4.1 (Geometric detection inside $\mathrm{Sp}^G_{\mathcal{O}}$)**.** *Let $Y \in \mathrm{Sp}^G_{\mathcal{O}}$ be geometric. Then for any integer n,*

$$Y \in \tau^{\mathcal{O},G}_{\geq n} \quad \Longleftrightarrow \quad \Phi^G(Y) \in \mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(G))}.$$

Proof. Consider the full subcategory $(\mathrm{Sp}^G_{\mathcal{O}})^{\mathrm{geom}} \subseteq \mathrm{Sp}^G_{\mathcal{O}}$ of geometric objects. Smashing with $\widetilde{E\mathcal{P}}_G$ is a smashing localization with essential image $(\mathrm{Sp}^G_{\mathcal{O}})^{\mathrm{geom}}$. In particular, Φ^G is conservative on $(\mathrm{Sp}^G_{\mathcal{O}})^{\mathrm{geom}}$.

Inside $(\mathrm{Sp}^G_{\mathcal{O}})^{\mathrm{geom}}$, the localizing subcategory $\tau^{\mathcal{O},G}_{\geq n} \cap (\mathrm{Sp}^G_{\mathcal{O}})^{\mathrm{geom}}$ is generated by those $\mathcal{O}$-slice cells that remain nonzero after smashing with $\widetilde{E\mathcal{P}}_G$. But any slice cell induced from a proper subgroup is annihilated by $\widetilde{E\mathcal{P}}_G$. Hence the only relevant generators have top subgroup G, and their G-geometric fixed points are the suspensions described in Lemma **??**(2). As in Proposition **??**, the least suspension degree appearing among generators of slice dimension $\geq n$ is $\mathrm{ceil}(n/w_{\mathcal{O}}(G))$. Therefore, restricted to geometric objects, Φ^G identifies $\tau^{\mathcal{O},G}_{\geq n} \cap (\mathrm{Sp}^G_{\mathcal{O}})^{\mathrm{geom}}$ with $\mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(G))}$. Conservativity of Φ^G on geometric objects yields the stated equivalence. □

5 Main theorem

Theorem 5.1 (Geometric fixed points characterize $\mathcal{O}$-slice connectivity)**.** *Let G be a finite group, $\mathcal{O}$ a transfer system on G, and let $X \in \mathrm{Sp}^G_{\mathcal{O}}$ be connective. Fix an integer $n \geq 0$. Then the following are equivalent:*

1. *X is $\mathcal{O}$-slice n-connective, i.e. $X \in \tau^{\mathcal{O},G}_{\geq n}$.*
2. *For every subgroup $H \leq G$, the geometric fixed points satisfy*

$$\Phi^H(\mathrm{Res}^G_H X) \in \mathrm{Sp}_{\geq \mathrm{ceil}\left(\frac{n}{w_{\mathcal{O}}(H)}\right)}.$$

Proof. We argue by induction on $|G|$.

(1)⇒(2). Assume $X \in \tau_{\geq n}^{\mathcal{O},G}$. Let $H \leq G$. By Lemma **??**, $\mathrm{Res}_H^G X \in \tau_{\geq n}^{\mathcal{O}|_H,H}$. Applying Proposition **??** in the group H gives $\Phi^H(\mathrm{Res}_H^G X) \in \mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(H))}$.

(2)⇒(1). Assume (2). Let $\mathcal{P}_G$ be the family of proper subgroups of G. Isotropy separation gives a cofiber sequence

$$E\mathcal{P}_{G+} \wedge X \longrightarrow X \longrightarrow \widetilde{E}\mathcal{P}_G \wedge X. \tag{1}$$

Since $\tau_{\geq n}^{\mathcal{O},G}$ is localizing, it suffices to show that both ends of (**??**) lie in $\tau_{\geq n}^{\mathcal{O},G}$.

Step 1: the proper-isotropy term. The G-CW complex $E\mathcal{P}_G$ is built from cells of the form G/H with $H < G$. Consequently $E\mathcal{P}_{G+} \wedge X$ lies in the localizing subcategory generated by the spectra $G/H_+ \wedge X \simeq \mathrm{Ind}_H^G \mathrm{Res}_H^G X$ for proper $H < G$. It therefore suffices to show that each $\mathrm{Ind}_H^G \mathrm{Res}_H^G X$ lies in $\tau_{\geq n}^{\mathcal{O},G}$.

Fix a proper $H < G$. By assumption (2), for every subgroup $K \leq H$ we have

$$\Phi^K\big(\mathrm{Res}_K^H \mathrm{Res}_H^G X\big) \cong \Phi^K(\mathrm{Res}_K^G X) \in \mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(K))}.$$

Since $|H| < |G|$, the induction hypothesis applies to the group H and the connective H-spectrum $\mathrm{Res}_H^G X$. It follows that $\mathrm{Res}_H^G X \in \tau_{\geq n}^{\mathcal{O}|_H,H}$. Then Lemma **??** implies $\mathrm{Ind}_H^G \mathrm{Res}_H^G X \in \tau_{\geq n}^{\mathcal{O},G}$. By localizing closure we conclude $E\mathcal{P}_{G+} \wedge X \in \tau_{\geq n}^{\mathcal{O},G}$.

Step 2: the geometric term. The spectrum $Y := \widetilde{E}\mathcal{P}_G \wedge X$ is geometric. Moreover $\Phi^G(Y) \simeq \Phi^G(X)$. By assumption (2) with $H = G$, we have $\Phi^G(Y) \in \mathrm{Sp}_{\geq \mathrm{ceil}(n/w_{\mathcal{O}}(G))}$. Lemma **??** therefore implies $Y \in \tau_{\geq n}^{\mathcal{O},G}$.

Finally, since both $E\mathcal{P}_{G+} \wedge X$ and $\widetilde{E}\mathcal{P}_G \wedge X$ are in $\tau_{\geq n}^{\mathcal{O},G}$, the middle term X is as well. □

Corollary 5.2 (Complete transfer systems recover Hill–Yarnall)**.** *If $\mathcal{O}$ is complete then $w_{\mathcal{O}}(H) = |H|$ for all $H \leq G$, and Theorem **??** specializes to the Hill–Yarnall characterization of the ordinary slice filtration: for connective X, $X \in \tau_{\geq n}^G$ if and only if $\Phi^H(\mathrm{Res}_H^G X) \in \mathrm{Sp}_{\geq \mathrm{ceil}(n/|H|)}$ for all subgroups H.*

References

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